COURSE SYLLABUS

Querying Microsoft SQL Server 2025

COURSE NUMBER	DURATION	AUDIENCE LEVEL	FORMAT
55317B	**3 days · 10 modules**	**Beginner**	**Instructor-Led / Self-Paced**

Course syllabus

COURSE OVERVIEW

55317B: Querying Microsoft SQL Server 2025 is a three-day beginner course that teaches students how to retrieve, analyze, and manage data using Transact-SQL and modern SQL Server tools. Students will use SQL Server Management Studio 22, Visual Studio Code, demonstrations, and hands-on labs to build practical querying skills for SQL Server 2025. This course provides a strong foundation for database administration, development, and business intelligence.

DELIVERY FORMAT OPTIONS

Standard Instructor-Led Training (ILT). The instructor teaches each module, leads explanations and discussions, and guides students through the standard module labs using the published module timing.

Bootcamp / Lab-Centered Delivery. When Bootcamp Mode capstone labs are included, they provide an alternative delivery configuration, not additional required ILT lecture time. In this configuration, students primarily work through module labs and capstone lab scenarios while the trainer acts as a facilitator, answering questions, clarifying concepts, and helping students work through lab challenges.

Training providers may choose either the normal ILT model or the Bootcamp/lab-centered model based on audience readiness, delivery goals, and available class time.

AT COURSE COMPLETION

After completing this course, you will be able to:

- Connect to, configure, and query SQL Server 2025 using SQL Server Management Studio 22 and Visual Studio Code.
- Author structured, efficient Transact-SQL queries to select, sort, filter, aggregate, and join relational database tables.
- Formulate safe data modification procedures utilizing transaction safety controls and structured error handling patterns.

Query and manipulate semi-structured XML and JSON payloads alongside native SQL Server 2025 AI-ready vector data.

Analyze query performance using graphical execution plans, indexes, and configure connections to external databases and file targets.

PREREQUISITES

- Basic understanding of relational database concepts
- Familiarity with the Windows operating system

COURSE OUTLINE

MODULE 1 Querying Data with SQL Server Management Studio (SSMS)

90 min

This module introduces students to the SQL Server 2025 database engine and teaches them how to navigate, configure, and execute queries using SQL Server Management Studio (SSMS) 22.

LESSONS

Lesson 1: Navigating the SSMS 22 Interface and Object Explorer

Lesson 2: Leveraging SSMS 22 Copilot and AI Query Assistants

Lesson 3: Leverage SSMS 22 Copilot And AI Assisted Tools

LAB EXERCISES

Connecting and Querying Database Engines in SSMS 22

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Connect to a SQL Server 2025 instance using SQL Server Management Studio 22
- Explore database objects and execute Transact-SQL scripts using the Query Editor
- Leverage SSMS 22 Copilot and AI-assisted tools to generate and explain queries
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 2 Querying Data with Visual Studio Code

90 min

This module covers the modern cross-platform development workflow using Visual Studio Code to execute queries, manage connection profiles, and format SQL code.

LESSONS

- **Lesson 1:** Managing SQL Connections and Script Files in VS Code
- **Lesson 2:** Manage SQL Connection Profiles Securely Inside VS Code Application
- **Lesson 3:** Write, Execute, And Analyze Transact SQL Scripts Using

LAB EXERCISES

- Writing and Executing Queries in VS Code

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure Visual Studio Code for local and enterprise Transact-SQL querying
- Manage SQL connection profiles securely inside VS Code settings
- Write, execute, and analyze Transact-SQL scripts using VS Code extensions
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 3 Basic Transact-SQL Queries

90 min

This module introduces students to the foundational elements of Transact-SQL (T-SQL) query structure, covering data retrieval, sorting, filtering, and handling NULL values.

LESSONS

- **Lesson 1:** Writing Foundational SELECT and WHERE Clauses
- **Lesson 2:** Sorting Results and Implementing NULL Logic
- **Lesson 3:** Sort Query Result Sets And Handle Logical NULL

LAB EXERCISES

- Writing Basic SELECT Statements

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Construct structured SELECT statements to retrieve columns from relational tables
- Filter output rows using conditional predicates in the WHERE clause
- Sort query result sets and handle logical NULL values correctly
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 4 **Advanced Transact-SQL Queries** 90 min

This module addresses complex querying patterns including table joins, aggregated grouping, scalar/correlated subqueries, and Common Table Expressions (CTEs).

LESSONS

- **Lesson 1:** Joining Relational Data Tables Safely
- **Lesson 2:** Aggregation, Subqueries, and CTEs
- **Lesson 3:** Draft Complex Queries Leveraging Subqueries And Common Table

LAB EXERCISES

- Implementing Joins and Subqueries

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Join multiple tables using INNER JOIN and LEFT/RIGHT OUTER JOIN techniques
- Group records and calculate aggregated values using GROUP BY and HAVING clauses
- Draft complex queries leveraging subqueries and Common Table Expressions
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 5 Optimizing Query Performance

120 min

This module covers fundamental concepts of query optimization in SQL Server 2025, focusing on execution plans, index design, and monitoring performance with Query Store.

LESSONS

- **Lesson 1:** Analyzing Graphical and Text-Based Execution Plans
- **Lesson 2:** Designing and Implementing Basic Indexes
- **Lesson 3:** Leveraging Query Store for Execution Analysis
- **Lesson 4:** Query SQL Server 2025 Using SQL Server Management

LAB EXERCISES

- Analyzing and Tuning Query Performance

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Interpret graphical execution plans to locate high-cost database operations
- Distinguish between clustered and non-clustered index types and apply performance enhancements
- Analyze query patterns using the built-in SQL Server Query Store diagnostic features
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 6 Modifying Data with INSERT, UPDATE, and DELETE

120 min

This module covers core techniques for modifying database tables, emphasizing standard insertion patterns, safe update clauses, deletions, and transaction safety.

LESSONS

- **Lesson 1:** Constructing Standard INSERT and Multi-Row Modification Statements
- **Lesson 2:** Executing Safe Updates and Deletes
- **Lesson 3:** Managing Transaction Blocks and Errors
- **Lesson 4:** Query SQL Server 2025 Using SQL Server Management

LAB EXERCISES

- Safe Data Modification and Transactions

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Construct INSERT, UPDATE, and DELETE statements while keeping constraints intact
- Formulate secure transaction blocks using BEGIN TRANSACTION, COMMIT, and ROLLBACK structures
- Incorporate defensive error handling in modification statements with TRY...CATCH blocks
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 7 Working with XML, JSON, and Vector (AI) Data

120 min

This module focuses on querying non-relational schemas, showing students how to leverage modern SQL Server 2025 structures for XML, JSON, and vector similarity models.

LESSONS

- **Lesson 1:** Parsing and Formatting JSON Payloads
- **Lesson 2:** Retrieving and Structuring XML Documents
- **Lesson 3:** SQL Server 2025 Native Vector Integration
- **Lesson 4:** Query SQL Server 2025 Using SQL Server Management

LAB EXERCISES

- Querying XML, JSON, and Vector Data

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Extract, format, and parse JSON and XML documents using Transact-SQL
- Query relational data and format relational results to nested JSON and XML files
- Store, index, and query vector data for semantic similarity lookups in SQL Server 2025
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 8 **Using Stored Procedures and Functions** 120 min

This module teaches students to compile SQL statements into Stored Procedures and reusable user-defined functions to encapsulate operational logic.

LESSONS

- **Lesson 1:** Creating Custom Parameterized Stored Procedures
- **Lesson 2:** Developing User-Defined Functions (UDFs)
- **Lesson 3:** Securing and Optimizing Code with Views
- **Lesson 4:** Query SQL Server 2025 Using SQL Server Management

LAB EXERCISES

- Creating and Executing Stored Procedures and Functions

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Author and modify custom parameterized Stored Procedures
- Design, write, and execute scalar and inline table-valued user-defined functions
- Construct code that passes values between procedures using variables and output parameters
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 9 **Querying SQL Server Programmatically** 120 min

This module explains how developers and administrators programmatic query SQL Server engines using PowerShell, drivers, and automation tooling.

LESSONS

- **Lesson 1:** Introduction to the SQL Server PowerShell Module
- **Lesson 2:** Automating Script Execution using Invoke-Sqlcmd
- **Lesson 3:** Parsing and Exporting SQL Results via Scripts
- **Lesson 4:** Query SQL Server 2025 Using SQL Server Management

LAB EXERCISES

- Querying SQL Server Programmatically with PowerShell

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Execute Transact-SQL queries programmatically using PowerShell scripts
- Utilize Invoke-Sqlcmd and the SQL Server powershell module safely
- Implement basic script parameters and handle relational arrays programmatically
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

MODULE 10 **Managing External Data Sources** 120 min

This module introduces methods to interact with databases, flat files, and servers outside of the immediate SQL Server 2025 instance using Linked Servers and PolyBase.

LESSONS

- **Lesson 1:** Establishing and Configuring Linked Servers
- **Lesson 2:** Querying Distributed Databases and Linked Instances
- **Lesson 3:** Accessing Flat Files with PolyBase Integration
- **Lesson 4:** Query SQL Server 2025 Using SQL Server Management

LAB EXERCISES

- Managing External Data Sources

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Establish secure connectivity to isolated database systems using Linked Servers
- Configure basic PolyBase schemas to read structured external files
- Write distributed queries combining local tables and external resources
- Query SQL Server 2025 using SQL Server Management Studio 22, Visual Studio Code, and modern Microsoft data tools

BOOTCAMP MODE — CAPSTONE LABS

Capstone labs are included for Bootcamp/lab-centered delivery. They are not additional required lecture time for Standard Instructor-Led Training. Use them when the client chooses a facilitated lab delivery model where the trainer coaches students through integrated practice scenarios.

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3, 4

SKILLS TO BE INTEGRATED

- Navigate and query database engines using SQL Server Management Studio 22 and VS Code
- Formulate basic Transact-SQL SELECT statements containing filter and sort clauses
- Construct multi-table JOIN operations and apply GROUP BY aggregation functions
- Author complex subqueries and Common Table Expressions to isolate targeted datasets

DAY 2 CAPSTONE Modules 5, 6, 7

SKILLS TO BE INTEGRATED

- Analyze execution plans and implement indexes to resolve slow-running SQL queries
- Draft and execute INSERT, UPDATE, and DELETE operations wrapped in safe, transactional blocks
- Generate and query XML, JSON, and high-dimensional vector embeddings with associated distance metrics

DAY 3 CAPSTONE Modules 8, 9, 10

SKILLS TO BE INTEGRATED

- Write and parameterize Stored Procedures and User-Defined Functions for reusable data operations
- Automate the extraction and manipulation of SQL data using PowerShell scripts and the SqlServer module
- Configure and query external relational or flat-file data sources using Linked Servers and PolyBase schemas

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.